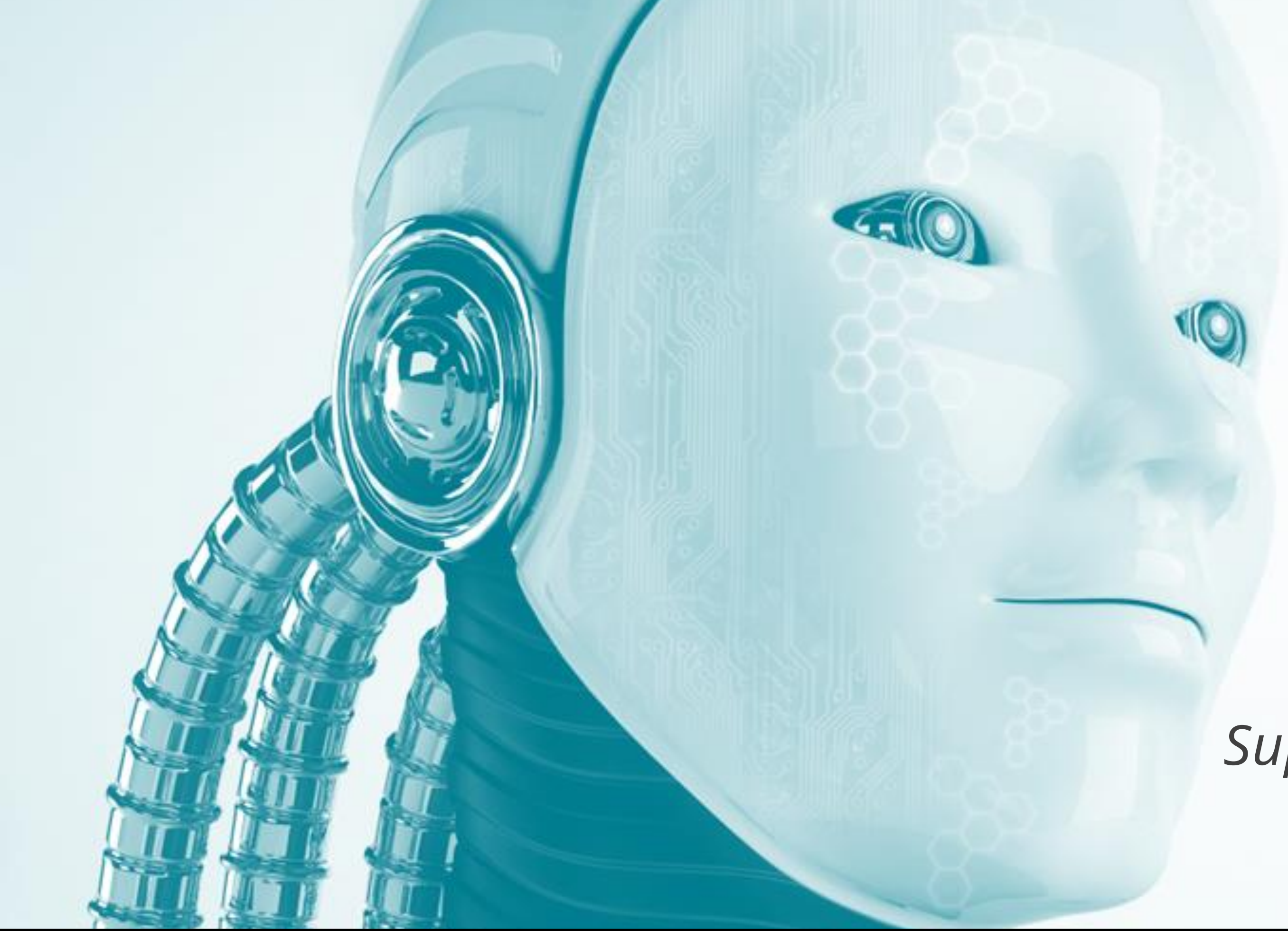


Contrastive Learning in Time Series

NorwAI

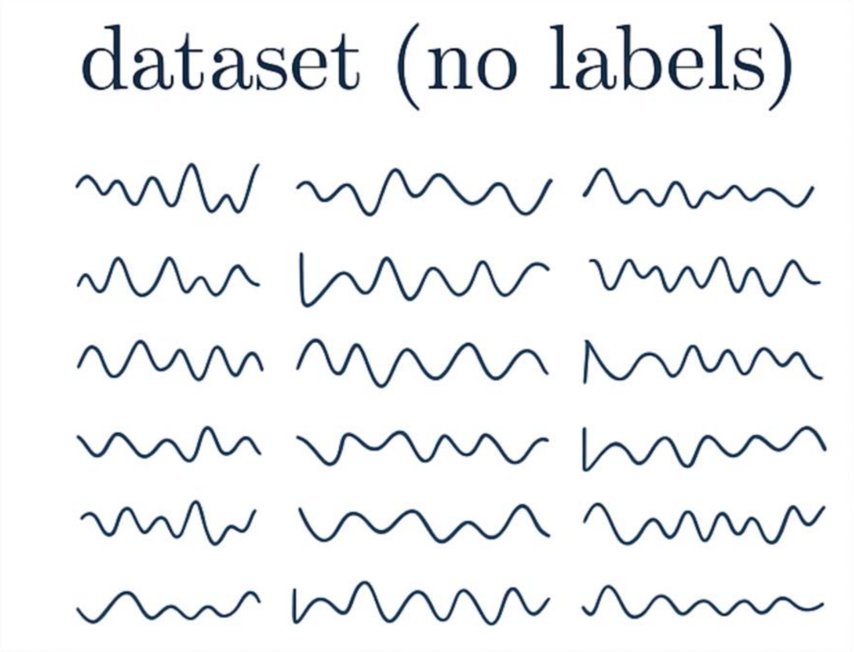
April 2026

Kazeem Shamba
Doctoral Researcher (PhD), NorwAI

Supervisors: Prof. Kerstin Bach & Gavin Taylor

Time Series / Sensor Data

Self-Supervised Learning Motivation



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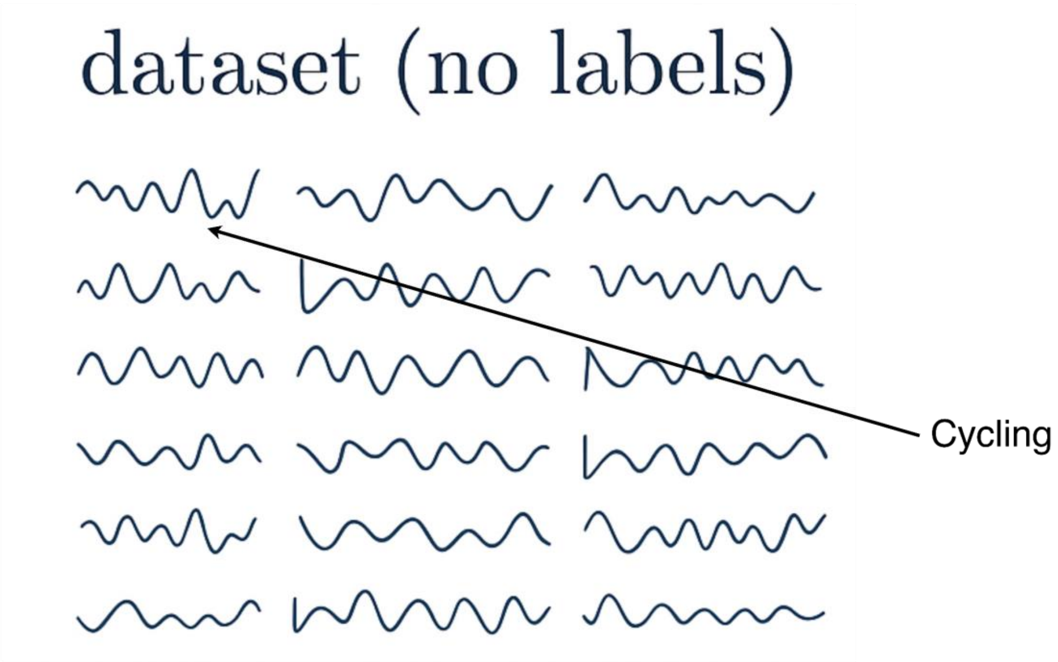

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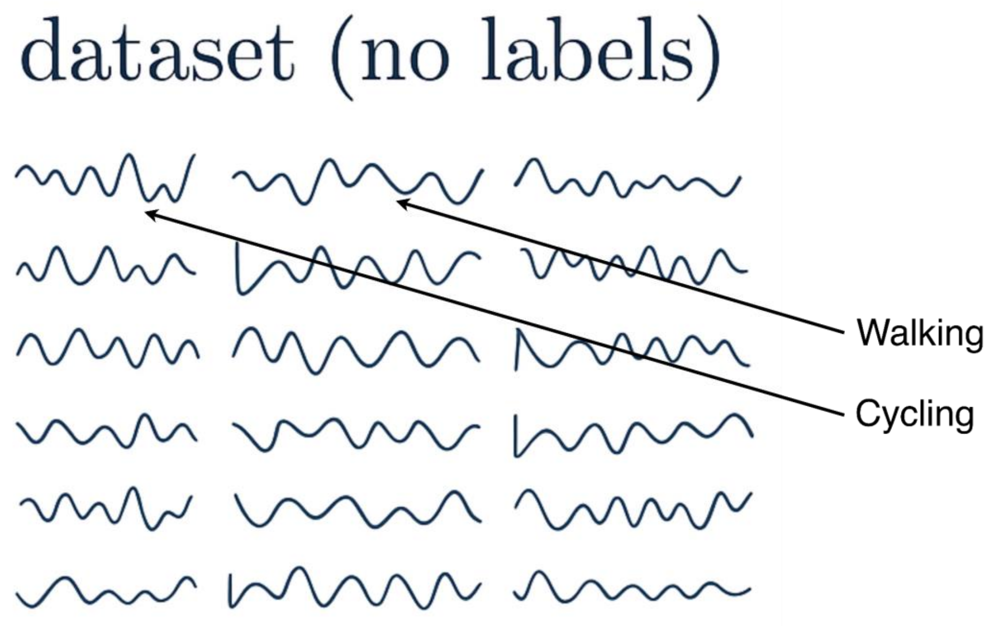

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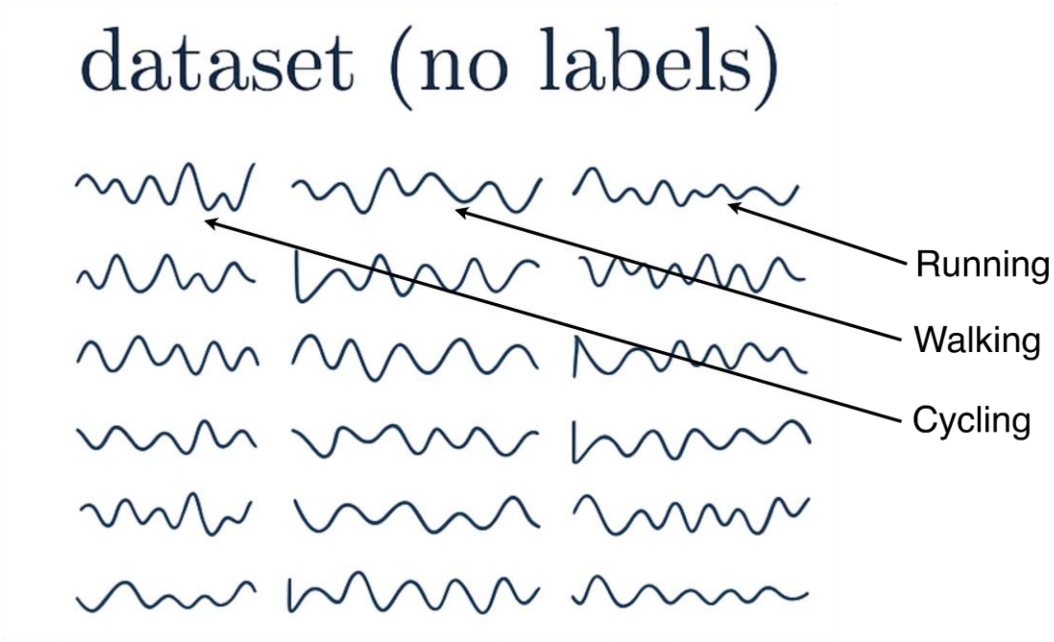
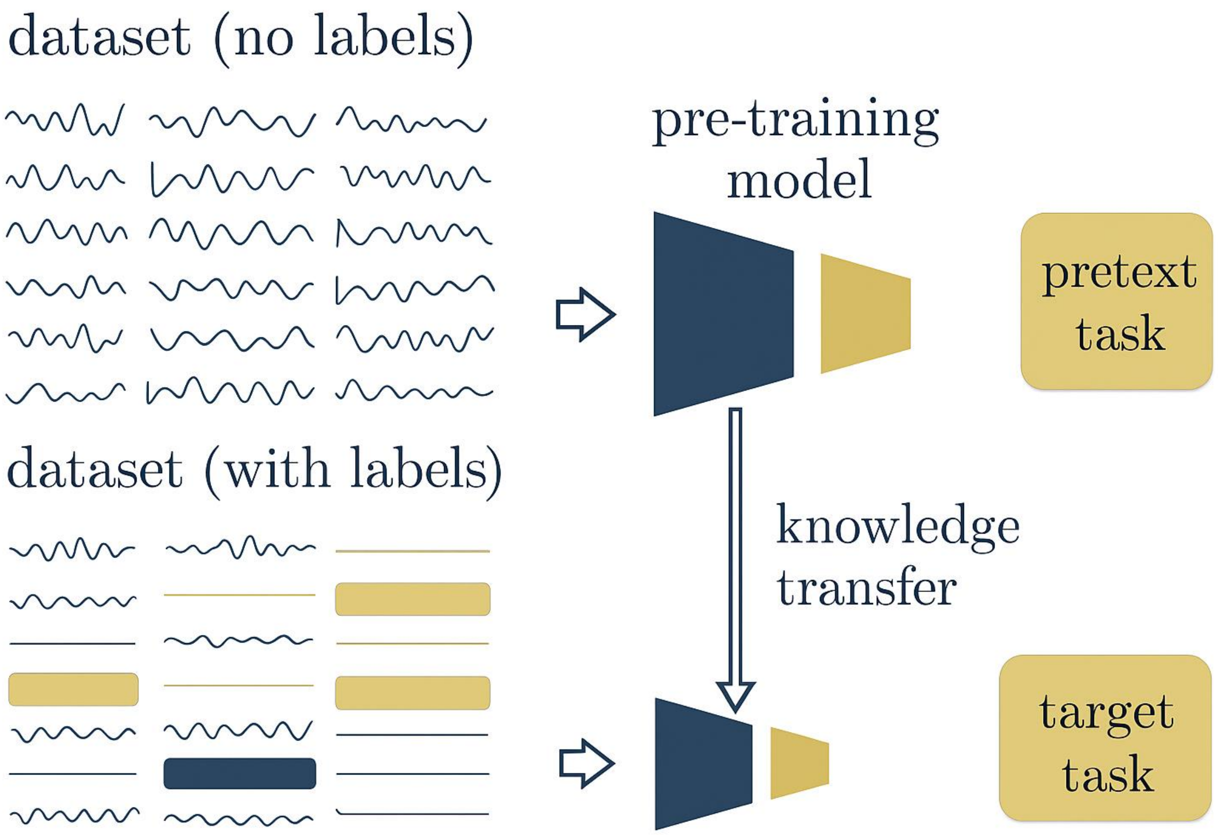


Supervised learning

- Relies on large labeled datasets
- Expensive and domain-specific annotations
- Limited generalization

Self-Supervised Learning Motivation



Supervised learning
- Relies on large labeled datasets
- Expensive and domain-specific annotations
- Limited generalization

Self-Supervised learning
- Learns representations directly from data
- Scalable (unlabeled data is abundant)
- Improves transferability to downstream tasks

Self-Supervised Learning – Pretext Task

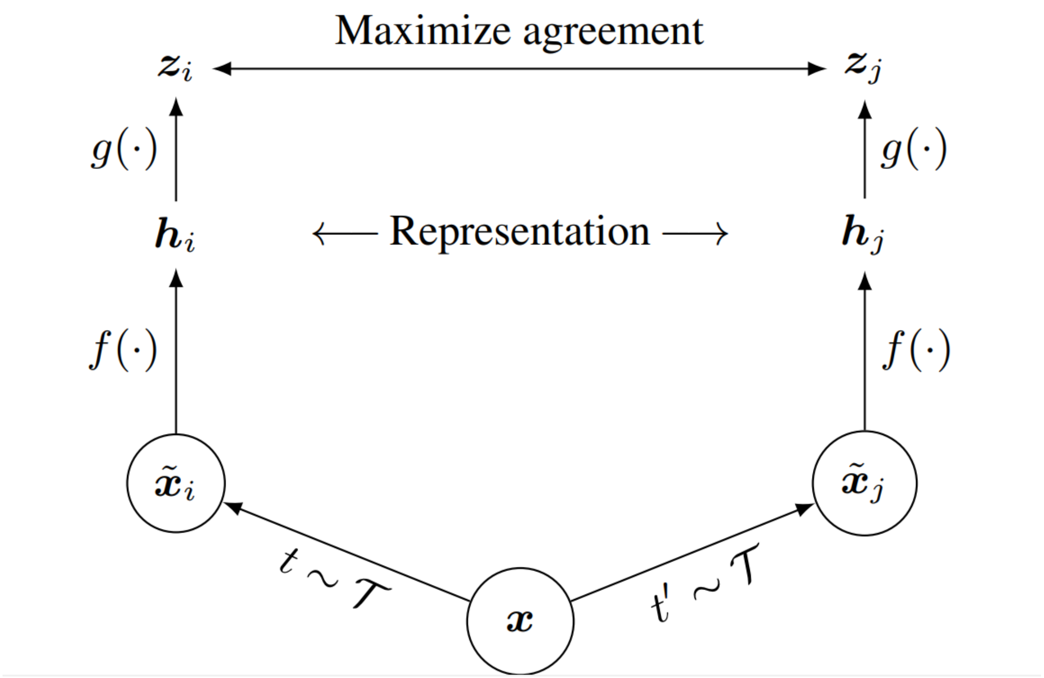
source: image generated by Microsoft copilot

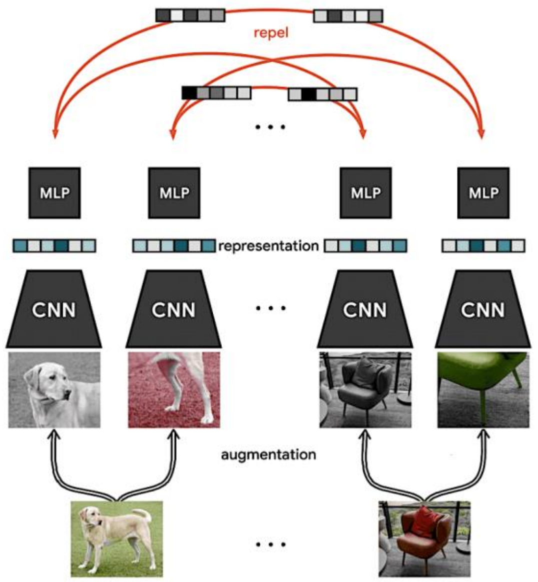
Pretext Tasks

- Jigsaw puzzle
- Inpainting
- Colorization
- Rotation
- Reconstruction

Contrastive Representation Learning

Computer Vision

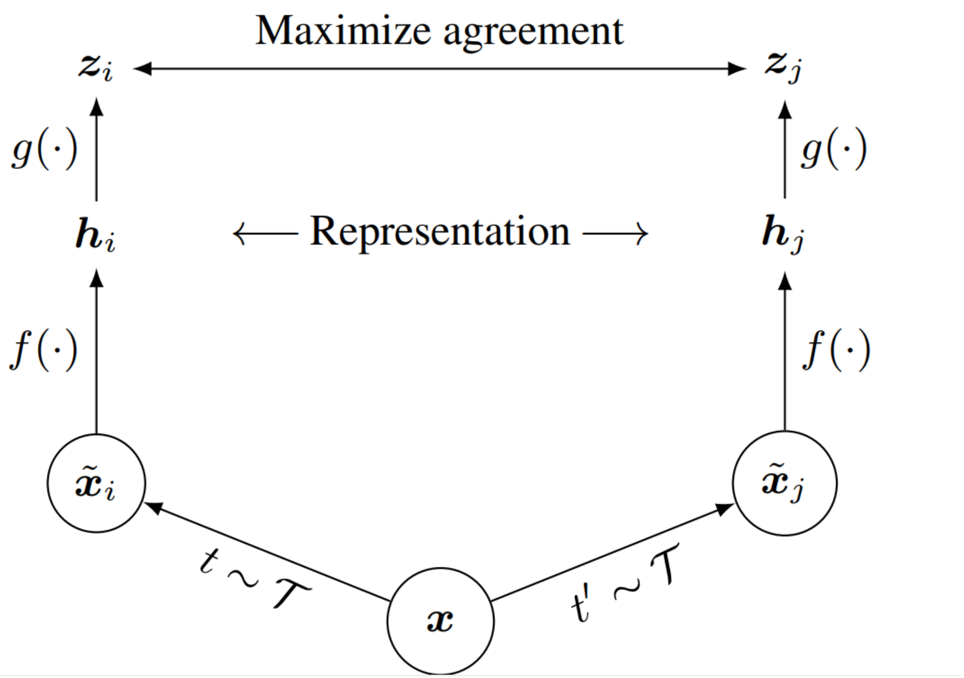


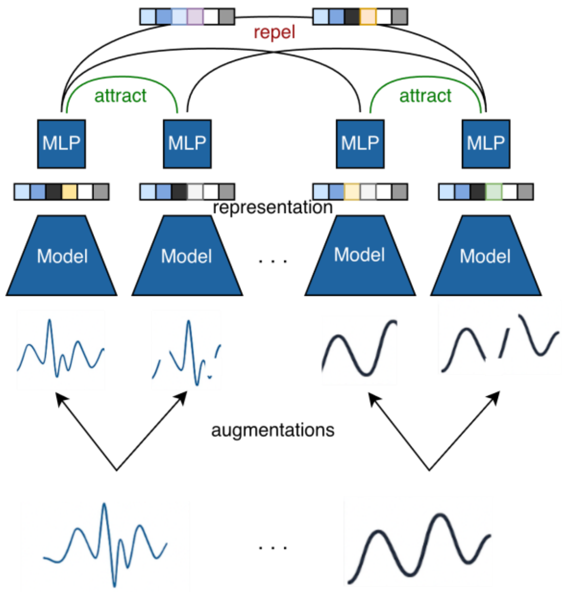


SimCLR, A Simple Framework for Contrastive Learning of Visual Representations, ICML 2020

Contrastive Representation Learning

Time Series

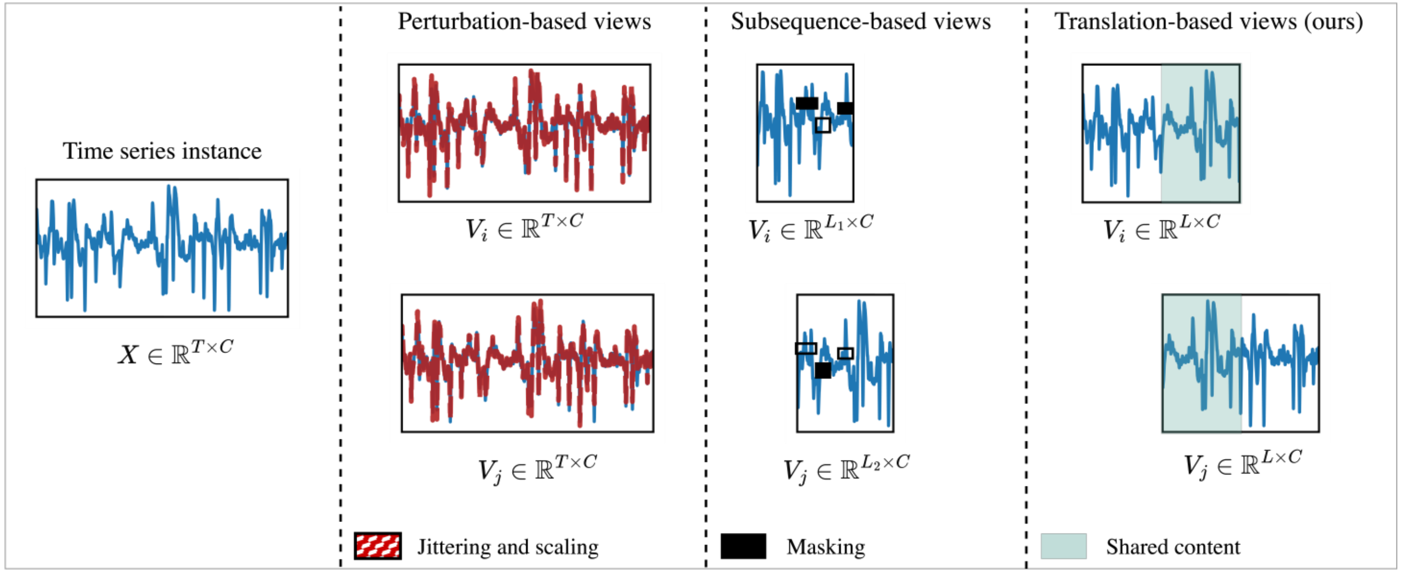




SimCLR, A Simple Framework for Contrastive Learning of Visual Representations, ICML 2020

Contrastive Learning in Time Series

- Augmentations and masking
- Multiple encoder passes for each view



Consequence

- Data distortion
- Scalability and speed

Contrastive Learning in Time Series

- *e*Margin, LDD Workshop ECAI, 2025
- Contrast All The Time (CaTT), Main Track ECAI, 2025
- Divide and Contrast (Di-COT), Main Track ICML 2026
- Learning by Shifting (ShiFT), ECML 2026 (*Under review*)
- Context Prediction (C-Pred), NeurIPS 2026 (*Under review*)

Success criteria

- No augmentation and less domain specific assumptions
- Improved accuracy and generalization
- Substantial reduction in training time
- Compact (clusters) and transferable representations

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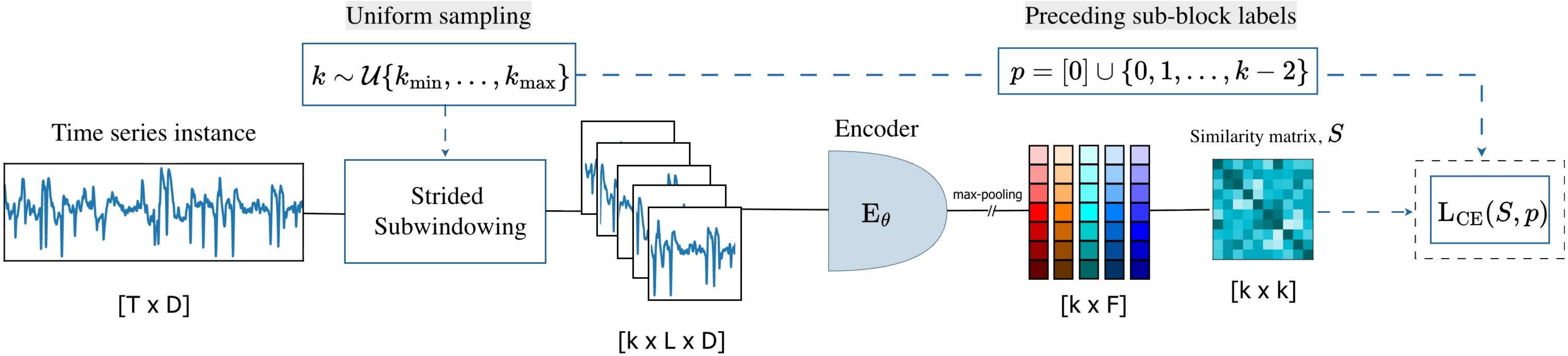
Divide and Contrast

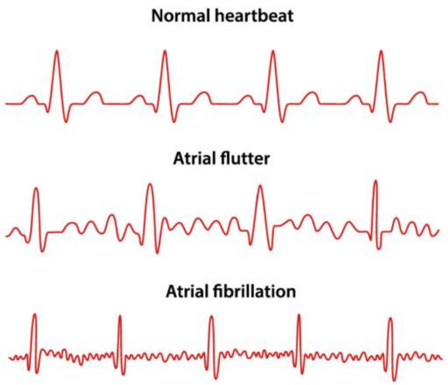
Learning Robust Temporal Features without Augmentation



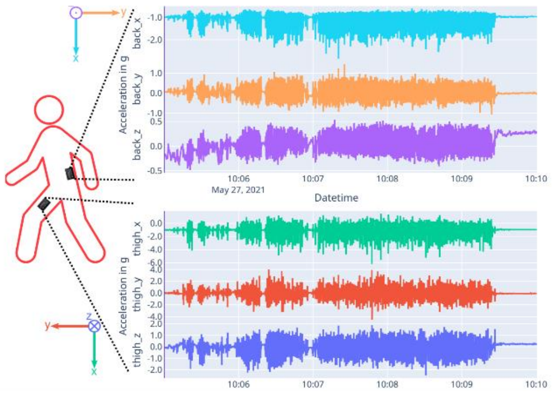
Divide and Contrast

Learning Robust Temporal Features without Augmentation

ECG

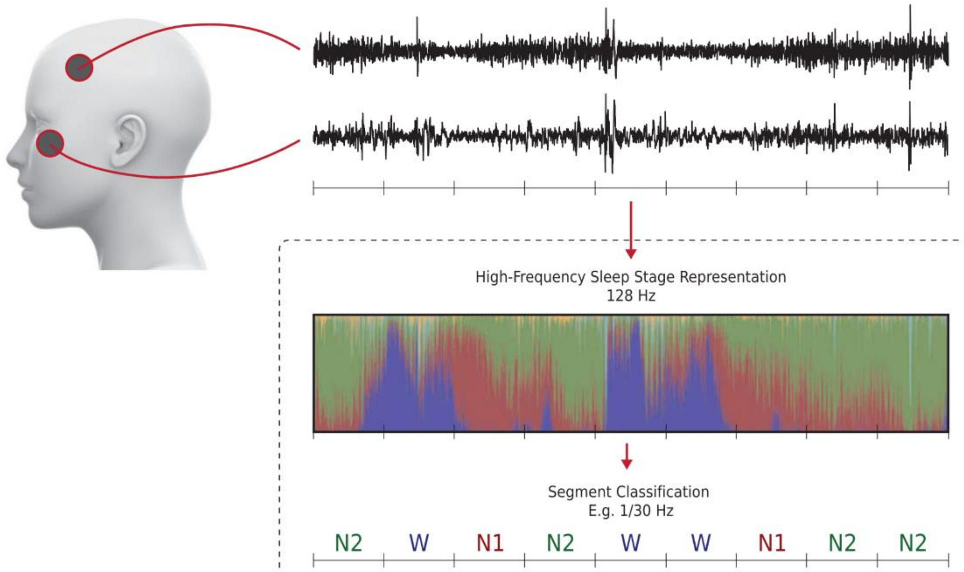


HARTH



Dataset	Train set	Test set	Subject	Sequence Length	Channel	Class
ECG	59,922	16,645	23	2,500	2	4
HARTH	23,025	2,785	22	500	6	18
PAMAP2	51,192	11,590	9	100	52	18
SKODA	22,587	5,646	1	50	64	11
SLEEP	25,612	8,910	20	3,000	1	5
WISDM2	134,614	14,421	29	40	3	6

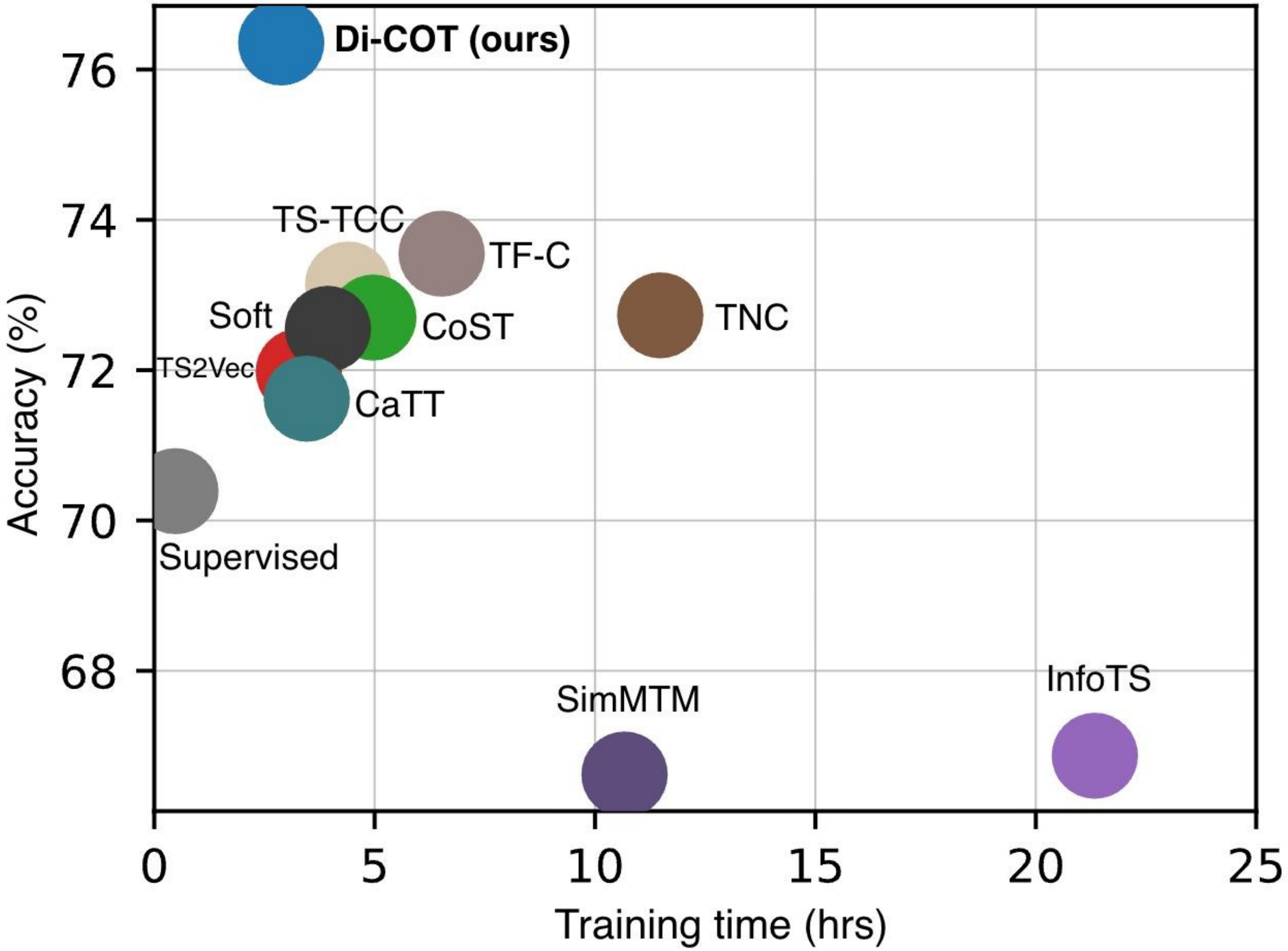
SleepEEG



source: https://ntnuopen.ntnu.no/ntnu-xmlui/handle/11250/3181556, https://norvinia.com/what-is-afib/, U-sleep dataset

Divide and Contrast

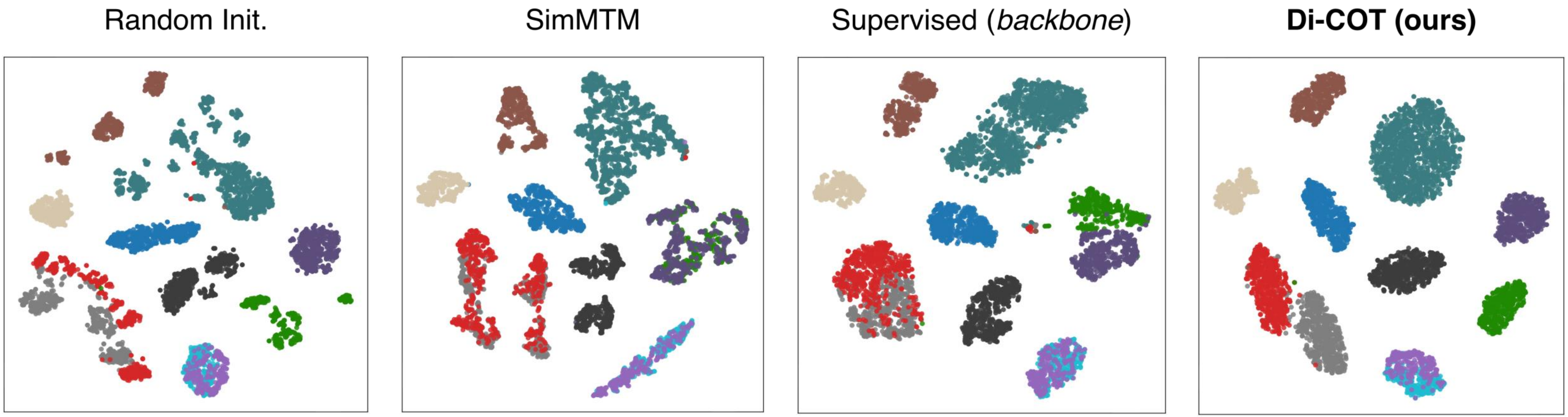
Learning Robust Temporal Features without Augmentation



- 1% of Labeled Data
- Highest Average Accuracy
- Fastest among self-supervised method

Divide and Contrast

Learning Robust Temporal Features without Augmentation



- Cleanly separates the challenging red and gray classes
- Supervised Backbone with access to full labeled train data (Limited generalization!)

- Translational invariance is *almost* all you need!
- When in doubt, just divide and contrast.

Read more!

Kazeem Shamba
Doctoral Researcher, NTNU

Abdulkks@ntnu.no